USB Type C Connector

J1 C393939 — GND, VBUS, CC1, DP1, DN1, SBU1, CC2, DP2, DN2, SBU2, VBUS, GND, SHELL ×4

R1 5.1k (CC1), R2 5.1k (CC2)

Set CC1 & CC2 to 5.1k
Pulls 5V 3A

D1 1n5819hw-7-f

C1 4.7uF

C34 1uF

U2 USBLC6-2SC6 — USB_D−, USB_D+

+3V3 Signal Bypass Caps

C48 1uF

VBUS

Battery

U8 TP4056 — VCC, CE, BAT, EP, TEMP, CHRG, STDBY, PROG, GND

VBUS

C21 10uF

C32 10uF

R19 1K, R21 1K

D2 GRN, D15 RED

R22 3K

BH-18650-A6AJ012

Q3 CJ3139K

TPVBAT+1 +BATT

R23 100 PWCC

C39 0.1uF

U11 DW01A — VCC, GND, OD, OC, CS

Q4 FS8205A_C2830320

R24 1K

+BATT

VBAT_RAW−

battery protection circuit

Rpg (k)	Ibat (mA)
10	130
5	250
4	300
3	400
2	580
1.66	690
1.5	780
1.33	900
1.2	1000

Power Switch

VBUS, VPWR_IN, +BATT

SW7

Battery to 3.3v Buck/Boost

U1 TPS63051RMWR — VIN, EN, ILIMO, GND, PG, SS, L1, GND, L2, VOUT, FB, PFM/PWM

VPWR_IN, +3V3

1.5uH L2

C40 10uF, C43 1nF

0 = PFM: Pwr Sav
1 = PWM

R25 1M, C44 10uF, C45 10uF

Input Current Limiter
ILM1/0

1	0	Current Limit
0	0	0.4x in
0	1	0.5x in
1	0	0.65x in
1	1	1.0x in

12MHz Crystal

ABM8-272-T3

Y1

15p C19, 15p C20

GND, GND

R11 1k

XIN, XOUT

IR Receiver

+3V3

R13 100

CQG1 IRM-H638T/TR2 — VCC, VOUT, GND, GND

GPIO19_IR_RECV

C2 4.7uF

GND

SWD Connector

SWD, SWCLK, +3V3, GND

J20 Conn_01x04_Pin

B/E Touch

GPIO26_ADC0_B — R16 1M — TOUCH_B

GPIO28_ADC2_E — R17 1M — TOUCH_E

Vibrator

+3V3

TP1 TestPoint

TP2 TestPoint

GPIO24

R32 100

Q6 AO3400A

R18 10k

GND

CTF LED

+3V3

D21 LED

R14

JP2 Jumper_2_Open

GPIO20_CTF_LED

R26 100

IR_LED_SIGNAL

Q1 AO3400A

R15 10k

C3 10uF

GND

Raspberry Pi RP2350A QFN-60

+1V1

C4 100n, C5 4.7uF, C6 100n

+3V3

C7 100n, C8 100n, C9 100n, C10 100n, C11 100n, C12 100n, C13 100n

R3 33, VREG_AVDD, C14 4.7uF

C15 4.7uF, C16 4.7uF, L1 3.3u

VREG_LX

USB_BOOT, R8 1k, SW1

SW2, R10 1k

RUN

SWCLK, SWD

XIN, XOUT

R4 27 USB_D+, R5 27 USB_D−

GPIO0	2	
GPIO1	3	GPIO1_BTN_A
GPIO2	4	GPIO2_BTN_L
GPIO3	5	GPIO3_BTN_R
GPIO4	7	GPIO4_1_1
GPIO5	8	GPIO5_1_2
GPIO6	9	GPIO6_1_SDA
GPIO7	10	GPIO7_1_SCL
GPIO8	12	
GPIO9	13	
GPIO10	14	GPIO10_0_1
GPIO11	15	GPIO11_0_2
GPIO12	16	GPIO12_0_SDA
GPIO13	17	GPIO13_0_SCL
GPIO14	18	
GPIO15	19	GPIO15_LED_DATA
GPIO16	27	
GPIO17	28	
GPIO18	29	
GPIO19	31	GPIO19_IR_RECV
GPIO20	32	GPIO20_CTF_LED
GPIO21	33	GPIO21_CHARGE_B
GPIO22	34	
GPIO23	35	GPIO23_CHARGE_E
GPIO24	36	
GPIO25	37	
GPIO26_ADC0	40	GPIO26_ADC0_B
GPIO27_ADC1	41	GPIO27_ADC1_VD Voltage Detect
GPIO28_ADC2	42	GPIO28_ADC2_E
GPIO29_ADC3	43	GPIO29_ADC3_VDE Voltage Detect Enable

Right SAO, Left SAO

U4

C18 10u

SAO

SAO 1.69bis

Pair of 22 caps to prevent spikes in power draw

SY6280A to prevent backchannel and overcurrent draw

U10 SY6280AAAC — IN, EN, OUT, GND, ISET

sao1_vcc

ISet A = 6800 / RSet
Min 0.4A / Max 2A

680mA = 10k

R20 10k

C37 10uF, C38 10uF

Simple_Addon_v2 J4 — VCC, SDA, GPIO1, GND, SCL, GPIO2

GPIO012_0_SDA, GPIO010_0_1, GPIO011_0_2, GPIO013_0_SCL

Simple_Addon_v2 J5 — VCC, SDA, GPIO1, GND, SCL, GPIO2

GPIO06_1_SDA, GPIO04_1_1, GPIO05_1_2, GPIO07_1_SCL

Push Buttons

SW11 GT-TC075B-H038-L1 — GPIO2_BTN_L — Left

SW12 GT-TC075B-H038-L1 — GPIO3_BTN_R — Right

SW10 GT-TC075B-H038-L1 — GPIO1_BTN_A — LOGO BTN

Voltage Detect

+BATT

R30 100k

Q5 AO3401A

Q2 AO3400A

R27 10k

4.7v = 2.35
3v = 1.5

GPIO27_ADC1_VD

GPIO29_ADC3_VDE

R29 100

R31 100k

R28 10k

C55 100n

Bling (LEDs)

+3V3

GPIO15, R12 220 LED Data

C36 0.1uF, C46 0.1uF, C22 0.1uF, C23 0.1uF, C24 0.1uF, C25 0.1uF, C26 0.1uF, C27 0.1uF

WS2812B — DI, VDD, GND, DO — D18, D19, D3, D4, D5, D6, D7, D8

LED0_1, LED1_2, LED2_3, LED3_4, LED4_5, LED5_6, LED6_7

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Title: Badgelife Village Badge 2026 – By Jeff "BigTaro"

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